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Adapting Swedish Integration Logics for Age-Friendly Retrofitting in China: Physical and Cognitive Perspectives

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Title	Adapting Swedish Integration Logics for Age-Friendly Retrofitting in China: Physical and Cognitive Perspectives
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Abstract

Population ageing is placing increasing pressure on housing to support ageing in place, particularly in environments that were not designed to accommodate changing functional needs over time. In China, age-friendly retrofitting remains fragmented, often focusing on isolated technical interventions rather than housing adaptation as an integrated system. This study develops a mechanism-oriented framework for understanding age-friendly retrofitting through the lens of Person–Environment (P–E) Fit theory, conceptualizing retrofitting as a process of recalibrating the relationship between human capability and environmental press. A comparative case approach is adopted, using Sweden as an analytical case to identify underlying mechanisms and China as a context for reinterpretation.

The findings reveal that effective retrofitting systems are based on two interrelated logics. The first is a physical integration logic, which coordinates regulatory standards, spatial design, and industrialized implementation to translate anthropometric knowledge into scalable spatial solutions. The second is a cognitive construction logic, through which functional design is framed, legitimized, and normalized in everyday life. In the Chinese context, these mechanisms remain fragmented across spatial, institutional, and socio-cultural dimensions, limiting both implementation and acceptance. The study argues that adaptation should not be understood as the transfer of a complete system, but as a process of reinterpreting and reconfiguring underlying mechanisms to local conditions. By shifting attention from design solutions to mechanism formation, this research provides a structured framework for understanding age-friendly retrofitting as a context-dependent and system-based process, offering theoretical and practical implications for ageing societies.

Titel	Anpassning av svenska integrationslogiker för äldrevänlig bostadsanpassning i Kina: fysiska och kognitiva perspektiv
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Sammanfattning

Befolkningens åldrande ställer ökade krav på bostäder att stödja kvarboende i hemmet, särskilt i miljöer som inte har utformats för att möta förändrade funktionella behov över tid. I Kina är äldrevänlig bostadsanpassning fortfarande fragmenterad och fokuserar ofta på enskilda tekniska åtgärder snarare än på bostadsanpassning som ett integrerat system. Denna studie utvecklar ett mekanismorienterat ramverk för att förstå äldrevänlig bostadsanpassning utifrån teorin om Person–Environment Fit (P–E Fit), där bostadsanpassning betraktas som en process för att omkalibrera relationen mellan människans förmåga och miljöns krav. Studien använder en komparativ fallstudieansats där Sverige fungerar som ett analytiskt fall för att identifiera underliggande mekanismer, medan Kina används som en kontext för vidare tolkning.

Resultaten visar att effektiva system för bostadsanpassning bygger på två sammanlänkade logiker. Den första är en fysisk integrationslogik som samordnar regelverk, rumslig utformning och industrialiserat genomförande för att omvandla antropometrisk kunskap till skalbara rumsliga lösningar. Den andra är en kognitiv konstruktionslogik genom vilken funktionell design ramas in, legitimeras och normaliseras i vardagslivet. I den kinesiska kontexten förblir dessa mekanismer fragmenterade över rumsliga, institutionella och sociokulturella dimensioner, vilket begränsar både genomförande och acceptans. Studien argumenterar för att anpassning inte bör förstås som en överföring av ett färdigt system, utan som en process där underliggande mekanismer omtolkas och omkonfigureras utifrån lokala förutsättningar. Genom att flytta fokus från enskilda designlösningar till hur mekanismer formas bidrar studien med ett strukturerat ramverk för att förstå äldrevänlig bostadsanpassning som en kontextberoende och systembaserad process, med både teoretiska och praktiska implikationer för åldrande samhällen.

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Stockholm, June 2026

Hao Pan

List of Abbreviations

ADL	Activities of Daily Living
HFI	Home Research Institute (Hemmens Forskningsinstitut)
IADL	Instrumental Activities of Daily Living
P–E Fit	Person–Environment Fit
SS 914221	Swedish Standard for Accessibility in Buildings
WHO	World Health Organization

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1 Introduction

1.1 Background

Population aging is a fundamental demographic shift reshaping the relationship between people and the built environment. As life expectancy increases and fertility declines, existing housing systems are increasingly misaligned with the needs of aging populations, making residential adaptation a critical global concern.

At the core of this challenge lies a structural mismatch between aging bodies and housing environments originally designed for younger, able-bodied occupants. In many countries, a significant share of the housing stock predates contemporary standards of accessibility, adaptability, and long-term usability. Consequently, common spatial configurations—such as narrow circulation routes, level differences, and compact service areas—often fail to meet the functional requirements of older adults.

Within this context, “aging in place” has emerged as a central policy objective and a key design challenge. Rather than relying on institutional care, many societies seek to enable older adults to remain in their homes for as long as possible. This shift places increasing pressure on existing housing systems, requiring not only technical modifications but also more systemic approaches to residential adaptation.

However, despite the global nature of this challenge, responses vary significantly across socio-spatial contexts.

1.2 Problem Statement

Within the broader challenge of adapting housing to aging populations, China represents a critical and highly complex context. The country is undergoing a demographic transition of unprecedented scale and speed. According to the National Bureau of Statistics of China (2023), the population aged 60 and above reached 297 million, accounting for 21.1% of the total population and marking the country’s transition into a deeply aging society.

This demographic shift exposes a fundamental mismatch between aging bodies and the existing housing stock. A large share of urban residential buildings in China was

constructed during the rapid urbanization of the late twentieth century, when design priorities emphasized efficiency and basic functionality rather than long-term adaptability (Qin, 2021). As a result, these dwellings often exhibit spatial configurations that are poorly aligned with the physiological and functional changes associated with aging.

Empirical studies consistently identify recurring spatial constraints, including stepped entrances, narrow circulation routes, level differences, and compact sanitary spaces (Zheng et al., 2024). While these issues are often framed as accessibility barriers, they reflect a deeper structural problem embedded in the organization of domestic space. In this sense, the mismatch is not merely technical but structural, revealing a lack of adaptability in the relationship between human capability and the built environment over time.

In response, “aging in place” has emerged as a key policy direction in China, aiming to enable older adults to remain at home while reducing pressure on institutional care systems. However, despite the introduction of national standards—such as the Code for Accessibility Design (GB 50763) and the Code for Design of Residential Buildings for the Aged (GB 50340)—implementation in existing housing remains limited and uneven.

In practice, interventions remain highly fragmented. Retrofitting typically focuses on discrete, localized modifications—such as handrails, ramps, or bathroom fixtures—without addressing the dwelling as an integrated spatial system. These “add-on” solutions often fail to address underlying issues of circulation, spatial hierarchy, and functional relationships, and may even introduce new discontinuities.

At the same time, physical fragmentation is reinforced by a fragmented cognitive environment. Age-friendly modifications are often associated with frailty and dependency, leading to hesitation, selective acceptance, or delayed adoption among residents (Wang et al., 2023). As a result, a persistent gap emerges between objective needs and subjective awareness, further constraining the effectiveness and scalability of retrofitting practices.

Taken together, these overlapping forms of fragmentation—both spatial and cognitive—undermine the development of a coherent approach to age-friendly retrofitting in China. Rather than functioning as an integrated process of spatial transformation, retrofitting remains a series of disconnected interventions, insufficient to respond to the evolving relationship between aging populations and the built

environment.

1.3 The Swedish Case

While the challenges of aging and housing adaptation are global, their responses vary significantly across contexts. In this study, Sweden is treated not as a normative model, but as an analytical case through which the underlying logics of age-friendly retrofitting can be systematically deconstructed.

Sweden's approach to aging in place is grounded in a long-term institutional and regulatory trajectory. Since the introduction of accessibility requirements in the 1970s—later incorporated into the Swedish Building Regulations—new housing has been required to meet standardized spatial criteria. At the same time, sustained efforts have been directed toward upgrading the existing housing stock, supported by detailed technical standards such as SS 914221:2006 (Building Design – Housing – Interior Dimensions) and public subsidy mechanisms.

Within this context, age-friendly retrofitting operates not as a series of isolated interventions but as a coordinated system linking regulatory frameworks, spatial design, and industrialized construction. Over time, these elements have become increasingly aligned, enabling retrofitting to be implemented in a consistent and scalable manner.

Equally important are the social and cognitive processes that shape this system. Accessibility and functional design have gradually been integrated into broader definitions of housing quality, allowing standardized spatial solutions to be widely accepted and embedded in everyday living environments.

As a result, Sweden represents a context in which both the physical organization and the cognitive acceptance of age-friendly retrofitting have developed in a relatively coherent manner. This coherence makes it possible to analytically deconstruct the mechanisms through which it is produced—specifically, how spatial standards, implementation systems, and social perceptions interact over time. This analytical positioning shifts the focus from outcome comparison to mechanism-based understanding.

1.4 Research Aim and Questions

This study develops a mechanism-based understanding of age-friendly retrofitting by examining contrasting conditions in China and Sweden.

Drawing on the Person–Environment (P–E) Fit framework, this study conceptualizes retrofitting as a process of reconfiguring the relationship between human capability and the built environment. Within this framework, particular attention is given to how this reconfiguration is achieved through both spatial systems and processes of social acceptance.

The analysis focuses on identifying and interpreting the underlying logics that structure age-friendly retrofitting practices. First, it deconstructs the Swedish system to identify its core operating principles, including the modular integration logic embedded in spatial standards and technical systems, and the cognitive construction logic underpinning their social acceptance. Second, it re-examines these principles within the Chinese context, exploring how they interact with different spatial, institutional, and cultural conditions, and what constraints and adaptation pathways emerge.

The research is guided by the following questions:

RQ1: What mechanisms structure the modular integration logic of age-friendly retrofitting in Sweden?
RQ2: Through what processes did accessibility standards in Sweden gain social acceptance and become embedded in everyday housing practices?
RQ3: How can these underlying logics be reinterpreted within the Chinese context, and what constraints and adaptation pathways emerge?

Sweden is therefore approached not as a blueprint, but as a heuristic case through which the interaction between spatial standards, institutional frameworks, and social acceptance can be analytically unpacked. This enables a more structured understanding of the challenges and possibilities of age-friendly retrofitting in China.

1.5 Thesis Structure

Chapter 2 reviews the literature on aging in place and residential retrofitting, identifying the key research gap addressed in this study. Chapter 3 establishes the theoretical framework based on Person–Environment Fit, linking spatial adaptation to the concept

of person–environment alignment. Chapter 4 outlines the research design and methodological approach.

Chapters 5 and 6 form the analytical core of the study, examining the physical integration logic and cognitive construction logic of the Swedish system. Chapter 7 reinterprets these logics within the Chinese context, identifying constraints and proposing an adaptive strategic framework. Chapter 8 concludes the thesis by summarizing the main findings, contributions, and limitations of the study.

2 Literature Review

2.1 Aging and Housing Mismatch in China

China is experiencing one of the most rapid demographic transitions globally, with the proportion of older adults increasing at an unprecedented scale. By the end of 2023, the population aged 60 and above reached 297 million (21.1%), while those aged 65 and above accounted for 217 million (15.4%), marking China's transition into a deeply aging society (National Bureau of Statistics of China, 2023).

Existing studies consistently highlight a fundamental mismatch between demographic aging and the characteristics of China's urban housing stock. A large share of residential buildings was constructed during the welfare housing period of the 1980s and the early commercial housing expansion of the 1990s, when design priorities emphasized efficiency and basic functionality rather than long-term adaptability (Qin, 2021).

Empirical studies consistently document recurring spatial constraints, including entrance level differences, narrow circulation paths, and compact bathroom layouts (Zheng et al., 2024). While these features are often framed as accessibility barriers, recent studies suggest that they reflect structural limitations embedded in the organization of domestic space rather than isolated design deficiencies. In this sense, the mismatch is structural, reflecting a lack of adaptability in the relationship between bodily capability and the built environment.

At the same time, the literature highlights a persistent gap between objective needs and subjective awareness. A survey of 5,485 older adults found that nearly 30% required home modifications, particularly safety-related interventions such as grab bars, yet overall awareness and proactive demand remained low (Wang et al., 2023). Moreover, longitudinal studies indicate that housing modifications are often undertaken as a "last resort," triggered by declining health rather than preventive planning (Wu et al., 2022). This gap between objective demand and subjective awareness constrains the timely adoption of retrofitting interventions.

Within this context, aging in place has emerged as a central policy strategy in response to rapid population aging and limited institutional care capacity. However, existing studies suggest that transforming dispersed and latent demand into a coordinated and scalable retrofitting system remains highly challenging, due to limitations in

standardization, implementation mechanisms, and social acceptance.

2.2 Development Differences in Age-Friendly Retrofitting

China and Sweden represent fundamentally different socio-economic and institutional contexts. Rather than attempting a broad comparison, this study focuses on a specific and comparable domain: the retrofitting of ordinary residential buildings to support aging in place. Within this bounded field, the comparison is structured as an examination of different developmental stages and underlying logics in age-friendly retrofitting.

Within this framework, Sweden can be characterized as operating in an "upgrade and maintenance" stage, while China remains in a "problem-solving and exploratory" stage. Sweden's system is characterized by established standards, stable demand, and coordinated implementation mechanisms, whereas China's system is still in formation, with evolving standards, market structures, and implementation practices.

In Sweden, system coherence is supported by three interrelated dimensions. First, a stable demand structure is supported by prevailing living patterns, in which most older adults remain in ordinary dwellings, often combined with home-based care services. This creates a consistent and large-scale basis for retrofitting interventions.

Second, a strong regulatory and technical foundation has been in place since the 1970s, resulting in a housing stock with baseline accessibility and supported by detailed spatial standards such as BFS 2024 and SS 914221:2006 (Boverket, 2024).

Third, these standards are embedded within coordinated policy and implementation frameworks, including national subsidy systems that enable systematic upgrading of existing housing. Large-scale projects, such as the Järva retrofitting programme, illustrate how existing housing can be incrementally aligned with contemporary requirements through coordinated interventions.

In contrast, China's development is characterized by partial and uneven progress across similar dimensions. While accessibility-related standards have been introduced, their application to existing housing remains limited and largely non-mandatory, resulting in the absence of a consistent technical baseline.

At the same time, a specialized retrofitting market remains underdeveloped, with

implementation largely driven by policy initiatives and pilot programmes. Without standardized solutions and mature market actors, implementation remains fragmented and difficult to scale.

This comparison highlights that the difference between Sweden and China is not merely one of outcomes, but of systemic maturity. This distinction provides the analytical foundation for the subsequent chapters, which focus on unpacking the mechanisms that structure these differing trajectories.

2.3 Physical Integration in Age-Friendly Retrofitting

Age-friendly retrofitting depends not only on individual technical solutions but on how design standards, spatial configurations, and construction processes are systematically coordinated. This coordination can be understood as a process of physical integration across multiple levels of the built environment.

In the Swedish context, this integration is historically rooted in the principles of Scientific Functionalism. Institutions such as the Home Research Institute (HFI) played a key role in translating ergonomic research and household efficiency into standardized spatial configurations. Over time, this approach led to the formation of an integrated system structured around three interrelated mechanisms:
(1) the codification of spatial requirements into design standards,
(2) the translation of these standards into standardized layouts, and
(3) their realization through prefabricated components.

Empirical studies show that such integrated systems enable retrofitting to be implemented in a predictable and scalable manner (Wu et al., 2022).

In contrast, studies on China predominantly describe a problem-oriented approach, in which interventions respond to specific safety risks such as fall prevention or barrier removal (An et al., 2025; Wang et al., 2023). While these measures address immediate functional needs, they are typically implemented as isolated modifications rather than as part of an integrated spatial system. The absence of detailed and enforceable standards, combined with a limited catalogue of standardized products, results in reliance on case-by-case design and on-site construction (Qin, 2021).

The literature therefore reveals a structural divergence in how physical retrofitting is organized. Integrated systems emphasize coordination across standards, design, and

production, whereas fragmented approaches operate through discrete interventions with limited scalability. This distinction provides the analytical basis for examining the mechanisms of physical integration in the subsequent analysis.

2.4 Cognitive Construction in Age-Friendly Retrofitting

The effectiveness of age-friendly retrofitting is shaped not only by technical and spatial factors, but also by cognitive processes that influence how interventions are perceived, interpreted, and accepted in everyday life. These processes determine whether spatial interventions are adopted, resisted, or delayed.

In the Swedish context, institutional actors played a key role in shaping the cognitive environment surrounding functional design. Through exhibitions, publications, and model environments, organizations such as the Home Research Institute (HFI) associated scientifically informed spatial design with broader societal values, including modernity, efficiency, and social progress.

This process can be understood as a cognitive construction logic comprising three interrelated mechanisms:
(1) the reframing of functional features as desirable design elements,
(2) the alignment of these features with broader social values, and
(3) their normalization through everyday use and experience.

Together, these mechanisms enabled accessibility features to be widely accepted and embedded within everyday housing practices.

In contrast, studies on China indicate a fragmented and less coherent cognitive environment surrounding age-friendly retrofitting. Age-friendly modifications are often associated with decline, dependency, or disability, which discourages proactive engagement. At the same time, there is no widely shared understanding of what constitutes an age-friendly living environment (Wang et al., 2023).

Housing adaptations are frequently undertaken only after significant functional decline, reflecting a reactive rather than preventive approach (Wu et al., 2022). This gap between objective need and subjective perception limits both the adoption and effectiveness of retrofitting interventions.

The literature therefore demonstrates that cognitive processes play a critical role in

shaping both the demand for and the acceptance of spatial interventions. The absence of a coherent narrative and shared value framework constrains the translation of technical solutions into widely adopted practices. This insight provides the foundation for analyzing cognitive construction mechanisms in the subsequent chapters.

2.5 Research Gap and Study Positioning

The existing literature reveals a fundamental tension in age-friendly retrofitting. On the one hand, there is increasing recognition of the need to adapt housing environments to aging populations, particularly in rapidly aging societies such as China. On the other hand, current practices remain constrained by fragmented implementation at the physical level and limited coherence in public perception at the cognitive level.

While studies on Sweden provide detailed accounts of policy frameworks, technical standards, and implementation practices, they tend to examine these elements in isolation. Limited attention has been given to the mechanisms through which physical systems and cognitive processes are interconnected and mutually reinforcing.

Conversely, research on China extensively documents practical challenges, including uneven standardization, limited market development, and low public awareness. However, these studies remain largely diagnostic, with limited attention to mechanism-based analysis or the development of structured and transferable frameworks.

This study addresses these limitations by adopting a mechanism-oriented perspective. It conceptualizes age-friendly retrofitting as shaped by two interrelated logics: a physical integration logic, involving the coordination of standards, design, and production, and a cognitive construction logic, involving processes of reframing, value alignment, and normalization.

Based on this framework, the study proceeds in two steps. First, it deconstructs the Swedish system to identify how these logics operate and interact. Second, it re-examines these logics within the Chinese context to explore their reinterpretation under different spatial, institutional, and cultural conditions.

This approach enables a shift from descriptive analysis to mechanism-based understanding, providing a structured basis for advancing age-friendly retrofitting in rapidly aging contexts. It also offers a framework for linking spatial design, institutional systems, and user perception within a unified analytical model.

3 Theoretical Framework

3.1 The Ecology of Aging

This study is grounded in Person–Environment (P–E) Fit theory, which conceptualizes the relationship between human capability and environmental demands as dynamic and interdependent.

The Ecology of Aging model conceptualizes the interaction between individuals and their environment through two key dimensions: individual competence, referring to physical, cognitive, and sensory capacities, and environmental press, referring to the demands imposed by the built environment. Optimal functioning occurs when competence and environmental press are balanced. As competence declines with aging, previously adequate environments may become increasingly demanding, leading to reduced independence and increased risk.

Within this framework, age-friendly retrofitting can be understood as an intervention that recalibrates environmental press to match changing human capabilities.

3.2 Retrofitting as Person–Environment Recalibration

Within this framework, the mismatch observed in China's housing stock can be understood as a structural imbalance between human capability and environmental demands. As populations age, individual competence declines, while the physical characteristics of existing housing remain largely unchanged. This results in an increasing mismatch, where environmental demands exceed the functional capacity of aging users.

This imbalance leads to spatial environments that no longer support everyday activities, increasing risks such as falls, mobility constraints, and functional dependency.

Age-friendly retrofitting functions as a process of recalibrating environmental press to restore person–environment fit. This involves reducing environmental press—such as eliminating level differences and widening circulation—and enhancing functional accessibility by improving reachability and maneuverability.

From this perspective, retrofitting is not merely a technical upgrade but a systematic

adjustment of the relationship between human capability and spatial environments.

3.3 Deriving the Dual-Logic Framework

The Person–Environment Fit framework provides the theoretical basis for deriving the dual-logic structure developed in this study. Specifically, it reveals that the adjustment of person–environment relationships operates through both material and perceptual dimensions.

• **Physical Integration Logic**

At the physical level, retrofitting operates by restructuring environmental press through spatial intervention. In the Swedish context, this restructuring is achieved through codified spatial standards, standardized layouts, and prefabricated systems. These mechanisms translate human requirements into spatial configurations, stabilizing the person–environment relationship at scale.

Anthropometry functions as an operational tool that quantifies human capabilities, enabling the calibration of environmental press in spatial design.

• **Cognitive Construction Logic**

In addition to physical adjustment, person–environment fit is also shaped by how environmental conditions are perceived and interpreted. Environmental press is not only objective but also subjectively experienced, varying with cultural meanings, social norms, and individual expectations.

This implies that achieving person–environment fit requires not only physical adjustment, but also the alignment of perception with spatial conditions.

In the Swedish context, functional design is framed as modern and desirable, accessibility features are normalized, and spatial changes are perceived as improvements rather than compromises. Cognitive construction thus operates by aligning perceived environmental conditions with acceptable and desirable levels of environmental press, facilitating the acceptance of standardized spatial solutions.

3.4 Repositioning Scientific Functionalism

Within this framework, Scientific Functionalism is understood not as a standalone

theory, but as a design paradigm that operationalizes person–environment fit.

Its emphasis on empirical observation, efficiency, and rational spatial organization provides a methodological basis for analyzing and optimizing environmental press. Through this approach, environmental demands are systematically measured, evaluated, and translated into spatial configurations.

Scientific Functionalism therefore serves as a methodological bridge between abstract theoretical principles and concrete spatial practices. It connects the principles of person–environment fit with practical mechanisms such as standardization, modular design, and prefabrication.

This connection explains how theoretical concepts of person–environment alignment are translated into systematic and scalable retrofitting practices.

4 Methodology

4.1 Research Design and Approach

This study employs a qualitative, mechanism-oriented research design to examine how age-friendly retrofitting systems are structured and how they operate across different contexts. Rather than comparing outcomes, the study focuses on analyzing underlying logics and their interaction.

The study is structured as a comparative case analysis of Sweden and China, focusing on differences in system development and underlying mechanisms. The comparison is not intended to establish equivalence, but to contrast different stages of system formation and integration.

The analysis proceeds in two stages. First, the Swedish system is deconstructed to identify the mechanisms that structure physical integration and cognitive construction. Second, these mechanisms are reinterpreted within the Chinese context to examine their applicability and constraints.

4.2 Case Selection

The selection of Sweden and China is guided by the need to examine contrasting configurations of age-friendly retrofitting systems. These two cases represent different stages of system development, allowing the analysis to focus on underlying mechanisms rather than contextual particularities.

Sweden is selected as a case of a highly integrated retrofitting system, characterized by established standards, stable demand structures, and coordinated implementation mechanisms. This makes it possible to identify how physical and cognitive mechanisms operate in a coherent and stabilized context.

China is selected as a case where age-friendly retrofitting systems are still in formation, characterized by fragmented implementation, evolving standards, and limited public awareness. This provides a context in which the constraints and adaptation pathways of these mechanisms can be examined.

The comparison is not intended to evaluate or rank the two contexts, but to construct

an analytical contrast between different stages of system development. This approach enables a mechanism-based understanding of how integration is achieved, rather than a descriptive comparison of institutional differences.

This case selection provides a structured basis for analyzing how physical integration and cognitive construction operate under different conditions, and how these mechanisms may be reinterpreted across contexts.

4.3 Data Collection

The study draws on multiple sources of documentary evidence to analyze the mechanisms underlying age-friendly retrofitting systems. Data collection is guided by the objective of identifying how physical integration and cognitive construction are structured and implemented.

The data can be grouped into three categories:
(1) policy and regulatory documents,
(2) technical standards and design guidelines, and
(3) academic and empirical studies.

Policy and regulatory documents are used to examine institutional frameworks, implementation mechanisms, and the role of governance in shaping retrofitting practices.

Technical standards, including the Swedish Building Regulations (BFS 2024) and SS 914221:2006, are analyzed to identify how spatial requirements are codified and translated into standardized design solutions.

Academic and empirical studies are used to interpret how these systems operate in practice, particularly in relation to spatial constraints, user behavior, and cognitive acceptance.

The selection of these data sources is guided by their relevance to the dual-logic framework, ensuring that both physical and cognitive dimensions of retrofitting can be systematically examined.

All data sources are drawn from publicly available and peer-reviewed materials, ensuring reliability and transparency. The combination of regulatory, technical, and empirical sources allows for a comprehensive analysis of both formal systems and practical implementation.

4.4 Data Analysis

The analysis adopts a qualitative, mechanism-oriented approach to identify how age-friendly retrofitting systems are structured and how they operate across contexts. Rather than describing individual cases, the analysis focuses on extracting underlying logics and their interaction.

The analysis is structured around the dual-logic framework developed in Chapter 3, distinguishing between physical integration and cognitive construction. These two dimensions guide both data interpretation and the identification of mechanisms.

The analytical process involves identifying patterns across different data sources and interpreting how they contribute to the formation of integrated or fragmented retrofitting systems. Particular attention is given to how spatial standards, design practices, and construction processes are coordinated, and how these are perceived and accepted by users.

The comparison between Sweden and China is conducted through an analytical contrast, examining how similar dimensions operate under different conditions of system development. This allows for the identification of both stable mechanisms and context-specific variations.

Through this process, the analysis seeks to move beyond descriptive accounts by identifying recurrent patterns that can be conceptualized as underlying mechanisms.

The findings are synthesized to explain how integrated retrofitting systems emerge, and how these mechanisms may be reinterpreted in different contexts.

5 Physical Integration Logic in Sweden

This chapter analyses how Sweden's age-friendly retrofitting system operates as a coordinated process linking regulatory standards, spatial design, and industrial production. Drawing on the Järva retrofitting project as a critical case, it traces how anthropometric knowledge is translated into codified requirements, standardized layouts, and prefabricated implementation.

Rather than consisting of isolated interventions, this process forms a structured system in which regulatory, design, and construction dimensions are tightly aligned. The following sections examine this system across three interdependent dimensions: regulatory foundation, design translation, and industrial implementation.

5.1 The Järva Project as a Critical Case

From a Person–Environment Fit perspective, large-scale retrofitting constitutes a systematic process of reducing environmental press embedded within existing housing structures. The Järva retrofitting project in Stockholm serves as a critical empirical case for examining how such processes are operationalized.

The Järva district represents a paradigmatic example of the Swedish Million Programme (1965–1974), which produced a large stock of standardized residential buildings prior to the introduction of mandatory accessibility regulations. As a result, these dwellings exhibit systematic spatial limitations, particularly in terms of vertical accessibility, constrained sanitary spaces, and limited adaptability. Previous research has shown that such post-war housing stock presents significant challenges for aging in place, especially in relation to mobility and accessibility (Slaug et al., 2020; Jonsson et al., 2021).

The retrofitting of Järva constitutes a coordinated intervention that addresses these structural deficiencies at scale. Accessibility improvements are implemented alongside energy upgrades and broader urban regeneration strategies, indicating that retrofitting operates as part of an integrated system rather than as isolated technical fixes.

A key example is the introduction of external prefabricated elevator systems to address vertical inaccessibility in mid-rise buildings. Rather than relying on extensive structural modifications, standardized lift shafts are attached to existing façades, enabling rapid

and repeatable installation across multiple buildings.

This approach transforms a building-specific constraint into a scalable solution, reflecting a shift from case-by-case intervention toward system-level standardization. From a Person–Environment Fit perspective, this intervention directly reduces environmental press associated with vertical mobility, thereby extending the functional capacity of residents with declining physical competence.

In addition, the upgrading of bathrooms—one of the most spatially constrained areas—demonstrates how accessibility requirements are translated into modular and spatially efficient systems. Swedish standards require sufficient space for wheelchair maneuverability (approximately a 1.5 m turning diameter), as well as adequate clearances and barrier-free floor conditions. Given the limited dimensions of original bathrooms, conventional on-site modification is both technically complex and highly disruptive.

To address this constraint, two primary strategies are adopted. The first involves the insertion of lightweight prefabricated bathroom units within the existing spatial envelope, using thin, high-performance materials to maximize usable space while meeting regulatory requirements. The second involves external expansion modules, in which additional bathroom volumes are attached to the building façade, increasing usable space without requiring major internal restructuring or resident relocation.

These approaches significantly reduce construction time and on-site disruption while enabling repeated application across similar housing units. In this sense, the intervention demonstrates how regulatory standards, spatial design, and industrial production are aligned to support scalable retrofitting processes.

These spatial reconfigurations reduce environmental press associated with movement, reachability, and daily activities, thereby enhancing functional independence and safety within the domestic environment.

The Järva project demonstrates that the effectiveness of age-friendly retrofitting lies not in isolated design solutions, but in the coordination of regulatory frameworks, spatial strategies, and industrialized implementation. It is this systemic alignment that enables retrofitting to operate as a scalable and reproducible process.

5.2 Regulatory Foundation

The effectiveness of Sweden's age-friendly retrofitting system is grounded in its regulatory framework, which translates anthropometric knowledge into codified and operational spatial norms.

At the core of this framework are the Swedish Building Regulations (BFS 2024) and design standards such as SS 914221:2006, which define spatial requirements based on empirical measurements of human dimensions, movement patterns, and assistive device use. These include specifications for door widths, circulation spaces, wheelchair turning radii, and functional clearances around fixtures.

Rather than relying on case-specific design interpretation, these standards establish a shared and measurable set of parameters that guide spatial configuration. Research has shown that such codification reduces ambiguity in decision-making and improves consistency across projects (Boverket, 2024; Slaug et al., 2020).

More importantly, anthropometric data enables the abstraction of human capability into quantifiable spatial requirements. This process transforms individual variability into standardized design criteria, allowing spatial environments to be systematically aligned with a wide range of user needs.

This transformation is central to scalability. Because spatial requirements are defined in fixed and measurable terms, they can be translated into repeatable design solutions and prefabricated components. Regulatory standards therefore function not only as guidelines but as enabling mechanisms that link design, production, and implementation.

From a Person–Environment Fit perspective, these codified norms operate as tools for calibrating environmental press. By defining spatial conditions that accommodate reduced mobility, limited reach, and assistive device use, they ensure that environmental demands remain within the functional capacity of aging individuals.

In this sense, the regulatory framework actively structures the relationship between human capability and the built environment. It provides a stable foundation upon which coordinated systems of design and construction can be developed.

5.3 Design Translation

The translation of regulatory standards into spatial design constitutes a critical intermediate mechanism through which age-friendly retrofitting becomes operational and scalable.

While regulatory frameworks define abstract spatial requirements, their implementation depends on how these requirements are organized within existing dwelling structures. In Sweden, this translation is achieved through standardized layouts and modular design systems, which convert codified norms into spatially coherent configurations.

A key enabling condition for this process is the typological consistency of the Million Programme housing stock. The repetition of similar unit layouts provides a stable framework within which recurring accessibility challenges—particularly in bathrooms, kitchens, and circulation areas—can be systematically addressed. Rather than designing each intervention independently, common spatial problems are resolved through predefined layout strategies.

These strategies integrate regulatory requirements—such as turning radii, clearance zones, and reachability criteria—into coordinated spatial configurations. As a result, isolated dimensional rules are transformed into functional spatial systems that support everyday activities.

From a Person–Environment Fit perspective, this process reorganizes environmental press at the level of spatial structure. Instead of addressing barriers individually, design translation reshapes the overall configuration of space, aligning movement paths and functional zones with the capabilities of aging users.

Standardized layouts also enable iterative refinement. As solutions are repeatedly applied across projects, they are progressively optimized in terms of spatial efficiency, constructability, and user experience. This process stabilizes design knowledge and increases predictability.

At the same time, modular design systems strengthen the link between spatial configuration and construction logic. By organizing space according to standardized components and dimensions, layouts can be directly translated into prefabricated elements.

In this sense, design translation functions as a bridge between regulatory abstraction and industrial implementation, ensuring that accessibility requirements are spatially integrated and practically achievable.

5.4 Industrial Implementation

Industrialized prefabrication constitutes the mechanism through which Sweden's retrofitting system achieves scalability, consistency, and practical implementation.

While regulatory standards define spatial requirements and design translation organizes them into layouts, prefabrication enables these configurations to be materialized in a controlled and repeatable manner. Industrial implementation thus links design intention with physical realization.

Standardized solutions—particularly in bathrooms and kitchens—are translated into prefabricated components engineered to meet regulatory specifications. Produced under controlled conditions, these components ensure precision and consistency across projects.

Prefabrication also minimizes on-site complexity. Instead of labor-intensive, case-by-case construction, prefabricated modules can be rapidly installed within existing dwellings. This reduces construction time, limits disruption, and improves efficiency (Boverket, 2024).

More importantly, prefabrication standardizes performance. Because components are based on fixed spatial parameters, the resulting environments exhibit consistent levels of accessibility, usability, and safety. This reduces variability and ensures reliable implementation of design intentions.

From a Person–Environment Fit perspective, this consistency stabilizes environmental press. It ensures that adapted environments remain within the functional capacity of aging users, supporting independence and reducing risk.

Prefabrication further reinforces the alignment between regulation, design, and construction. Standards define dimensions, layouts organize them spatially, and prefabricated components materialize them directly. This tight coupling transforms retrofitting into an integrated system.

Taken together, prefabrication does not merely improve efficiency; it enables systemic

integration. It is through this industrial logic that retrofitting becomes scalable, reproducible, and capable of addressing large housing stocks in a consistent manner

6 Cognitive Construction Logic in Sweden

This chapter examines how Sweden's age-friendly retrofitting system operates at the cognitive level, focusing on the processes through which spatial configurations are perceived, interpreted, and accepted. While Chapter 5 analysed how environmental press is structured through regulatory, spatial, and industrial mechanisms, this chapter traces how these configurations become cognitively stabilized.

Rather than treating cognition as an abstract or purely individual factor, the analysis considers it as a structured process shaped by institutional practices, design strategies, and repeated interaction with the built environment. The following sections examine this process through the role of institutional actors, the alignment between spatial design and social values, and the normalization of spatial configurations through experience.

6.1 HFI and the Reframing of the Domestic Environment

This section examines the role of institutional actors—most notably the Home Research Institute (HFI)—in shaping Sweden's cognitive construction logic by redefining how domestic space was understood, evaluated, and designed.

HFI redefined the home as a domain of scientific analysis rather than a static or tradition-bound environment. Through time–motion studies, ergonomic analysis, and systematic observation, everyday domestic activities—such as cooking, cleaning, and movement—were interpreted as measurable and optimizable processes.

This shift directly restructured spatial design. Layout decisions—including the organization of kitchen work zones, circulation paths, and spatial relationships between rooms—were increasingly informed by quantified patterns of use rather than convention or personal preference.

This transformation extended beyond design practice to reshape perception. By framing spatial organization as a matter of scientific optimization, HFI redefined what constituted a "good" home. Features such as wider circulation, simplified layouts, and standardized configurations were no longer perceived as constraints, but as rational and desirable improvements.

From a Person–Environment Fit perspective, this shift represents a transformation in how environmental press is perceived. Spatial constraints were no longer understood as

fixed conditions, but as adjustable variables that could be optimized to better match human capabilities.

This reframing reduced resistance to spatial change. Design interventions that might otherwise have been perceived as intrusive or restrictive—such as layout reorganization or dimensional standardization—became acceptable as they were associated with improved functionality and everyday efficiency.

In this sense, HFI did not merely produce technical knowledge, but actively constructed a cognitive framework through which spatial design was interpreted and evaluated. It established the foundation for a broader cultural shift in which functional and accessible environments were understood as integral to modern living.

This process demonstrates that the effectiveness of age-friendly retrofitting depends not only on physical interventions, but also on how these interventions are cognitively framed and socially understood.

6.2 Mechanisms of Cognitive Construction

This section examines three mechanisms through which cognitive construction is structured within Sweden's retrofitting system: methodological framing, narrative alignment, and experiential normalization.

The first mechanism operates through methodological framing. By grounding spatial design in empirical observation and scientific analysis, domestic environments are positioned as objects of rational optimization rather than personal preference. This reframing establishes a cognitive basis in which spatial configurations are evaluated according to measurable performance criteria, making functional design appear objective, necessary, and legitimate.

The second mechanism operates through narrative alignment. Policy discourse, design guidelines, and broader modernization narratives collectively frame accessibility and functional efficiency as desirable social values. Through this alignment, spatial adaptations are not perceived as technical adjustments, but as part of a broader progression toward modern, rational, and inclusive living environments.

The third mechanism operates through experiential normalization. As residents interact with modified environments over time, spatial configurations that initially appear

unfamiliar or restrictive become normalized through repeated use. This process stabilizes perception by aligning lived experience with designed spatial conditions, reinforcing acceptance and reducing cognitive resistance.

Together, these mechanisms structure how environmental conditions are perceived, interpreted, and accepted. Cognitive construction is therefore not an individual or spontaneous process, but a systemically produced outcome shaped by institutional practices, design strategies, and everyday interaction.

In this sense, cognitive construction operates as a complementary dimension to physical integration. While spatial interventions reduce environmental press in objective terms, cognitive mechanisms ensure that these changes are perceived as appropriate, acceptable, and desirable. The effectiveness of retrofitting thus depends on the alignment between material conditions and their cognitive interpretation.

6.3 Integrating Physical and Cognitive Mechanisms

The analysis of Sweden's retrofitting system reveals that physical integration and cognitive construction operate as interdependent mechanisms rather than separate dimensions.

Physical integration defines how environmental press is structured and reduced through regulatory standards, spatial configurations, and construction systems. Cognitive construction, in turn, shapes how these spatial conditions are perceived, interpreted, and accepted in everyday use.

These two dimensions interact in a mutually reinforcing manner. Standardized spatial configurations enable consistent user experience, while cognitive acceptance stabilizes the use of these configurations over time. This interaction ensures that retrofitting outcomes are not only technically effective but also socially and functionally sustainable.

As a result, retrofitting operates as a coordinated system in which regulatory frameworks, design practices, industrial production, and user perception are tightly aligned. This alignment reduces fragmentation across different stages of the process and enables consistent implementation at scale.

The Swedish case demonstrates that the effectiveness of age-friendly retrofitting does

not depend solely on technical innovation, but on the alignment between material interventions and cognitive processes.

This integrated perspective provides a basis for examining how similar mechanisms may operate under different conditions. In contexts where regulatory systems are less developed or cognitive alignment is weaker, the absence of such coordination may limit the effectiveness and scalability of retrofitting interventions.

7 Contextual Adaptation: A Strategic Framework for China

7.1 Fragmentation as a Condition of Mechanism Disruption

Building on the dual-logics framework developed in Chapters 5 and 6, the Chinese context can be understood as a condition in which the mechanisms of physical integration and cognitive construction are only partially formed and unevenly connected. Rather than representing isolated contextual differences, the conditions observed in China reflect a systemic fragmentation that disrupts the formation, coordination, and stabilization of these mechanisms.

This fragmentation does not imply the absence of technical solutions or policy initiatives. Instead, it indicates that the underlying processes linking regulation, design, production, and perception remain discontinuous. As a result, retrofitting interventions tend to operate as localized adjustments rather than as components of an integrated system.

• Physical and Built Environment Constraints

China's existing housing stock presents conditions that directly disrupt the mechanisms of design translation and industrial implementation identified in the Swedish system. The prevalence of high-density and highly heterogeneous residential environments limits the applicability of standardized spatial solutions.

Unlike the relatively uniform typologies of the Swedish Million Programme, Chinese housing encompasses a wide range of structural systems, unit layouts, and construction qualities. This heterogeneity prevents the stabilization of repeatable layout strategies, weakening the mechanism through which regulatory requirements can be translated into coherent spatial configurations.

At the same time, structural constraints—such as limited shaft space, rigid load-bearing systems, and constrained unit dimensions—restrict the feasibility of prefabricated insertion and modular expansion. As a result, the industrial implementation logic observed in Sweden cannot be effectively reproduced.

From a mechanism-oriented perspective, this condition reflects not merely a technical challenge, but a disruption in the linkage between standardized design and scalable construction. The tension between the need for large-scale retrofitting and the reality of

spatial diversity further reinforces this disruption, making system-level integration difficult to achieve.

• Socio-Economic and Governance Constraints

Beyond spatial conditions, the institutional structure of China's housing system constrains the formation of coordinated regulatory and implementation mechanisms. The fragmentation of property ownership, with most dwellings privately owned, fundamentally limits the possibility of building-level or system-wide retrofitting interventions.

Unlike contexts with centralized housing management, retrofitting decisions in China are distributed across individual households, resulting in significant variation in willingness, financial capacity, and timing. This condition prevents the alignment of interventions across multiple units, weakening the coordination necessary for systemic implementation.

In addition, the retrofitting market and industrial supply chain remain underdeveloped. The absence of standardized products, integrated service providers, and mature production systems limits the translation of design solutions into scalable construction practices. As a result, interventions rely heavily on case-by-case, on-site construction, introducing variability and reducing efficiency.

Furthermore, multi-level governance structures generate inconsistencies in policy interpretation and implementation across regions. From a mechanism perspective, these conditions reflect a weak coupling between regulatory frameworks, market systems, and construction processes, preventing the emergence of a stable and reproducible retrofitting system.

• Socio-Cultural and Cognitive Constraints

Cognitive and cultural factors further constrain the development of age-friendly retrofitting by limiting the formation of shared perception frameworks. Age-friendly modifications are often associated with decline, dependency, or disability, which discourages proactive engagement and delays intervention.

From a cognitive construction perspective, this condition reflects the absence of mechanisms that reframe spatial interventions as desirable and socially valued. Unlike the Swedish context, where accessibility features are integrated into broader narratives of modernity and quality of life, the Chinese context lacks a coherent value alignment

that supports the acceptance of such interventions.

As a result, spatial modifications are frequently minimized, concealed, or postponed. Even when objectively necessary, interventions may be perceived as undesirable or stigmatizing, creating a gap between environmental needs and user acceptance.

In addition, family-based decision-making introduces multiple layers of negotiation, further complicating the alignment between spatial intervention and user perception. This reinforces a tendency toward incremental and reversible modifications, rather than comprehensive spatial transformation.

From a mechanism-oriented perspective, these conditions indicate that the processes of narrative alignment and experiential normalization remain underdeveloped, limiting the stabilization of cognitive acceptance in retrofitting practices.

Taken together, these constraints indicate that the challenge of age-friendly retrofitting in China lies not in the absence of individual solutions, but in the disruption and incomplete formation of the mechanisms that enable their integration.

From a dual-logics perspective, both physical integration and cognitive construction remain fragmented, with weak linkages across regulatory frameworks, spatial design, industrial production, and user perception.

This condition establishes the boundary within which any adaptation of the Swedish logics must operate. It suggests that the key task is not to replicate an integrated system, but to reconfigure its underlying mechanisms in ways that accommodate heterogeneity, partial implementation, and evolving cognitive acceptance.

7.2 Reconfiguring Physical Integration Mechanisms

Physical integration in the Chinese context does not operate through replication of the Swedish model, but through the reconfiguration of underlying mechanisms under conditions of heterogeneity and limited coordination.

Rather than functioning as a fully integrated system, physical integration is characterized by partial and adaptive configurations in which regulatory frameworks, spatial design, and construction practices are reassembled in context-dependent ways.

• Regulatory Codification

In contrast to the standardized regulatory framework observed in Sweden, the Chinese context requires a more flexible approach to the codification of spatial requirements. The diversity of building types and structural conditions limits the applicability of uniform standards, reducing the effectiveness of rigid specification.

From a mechanism perspective, this reflects a shift in how regulatory frameworks operate. Rather than defining fixed spatial parameters, regulations function as adaptable reference systems that guide decision-making across varying conditions.

This transition redefines codification as conditional and context-sensitive, allowing regulatory principles to be maintained while enabling variation in spatial implementation.

• Design Translation

The mechanism of design translation, which in Sweden relies on typological consistency and standardized layouts, is fundamentally constrained by spatial heterogeneity in the Chinese context. The absence of repeatable unit types prevents the stabilization of universal layout strategies.

Design translation therefore shifts from standardized configuration to adaptive spatial negotiation. Spatial solutions are developed through the selective adjustment of existing structures, often within tight physical constraints.

This shift reduces scalability but enables localized alignment between spatial conditions and user needs.

• Industrial Implementation

The industrial implementation mechanism observed in Sweden, characterized by prefabrication and standardized production, is only partially transferable. Structural constraints, fragmented supply chains, and limited system coordination restrict the widespread adoption of prefabricated systems.

Construction processes therefore remain dependent on on-site, case-specific interventions, increasing variability and limiting efficiency.

From a mechanism perspective, industrial implementation shifts toward hybrid configurations in which prefabricated elements, when feasible, are combined with conventional construction practices.

Taken together, physical integration in China operates through partial alignment across regulatory, spatial, and construction dimensions rather than through a fully coordinated system.

Its effectiveness depends not on the replication of standardized systems, but on the capacity to reconfigure underlying mechanisms in response to contextual variability.

Table 1: Reconfiguration of Physical Integration Mechanisms in the Chinese Context

Mechanism Dimension	**Swedish System (Integrated Model)**	**Constraints in the Chinese Context**	**Mechanism Reconfiguration**
Regulatory Codification	Standardized spatial requirements derived from anthropometric data (e.g., BFS 2024, SS 914221)	Weak applicability of uniform standards under diverse building and structural conditions	Conditional and adaptive codification, where regulatory principles guide context-specific spatial decisions
Design Translation	Standardized layouts based on typological consistency enable repeatable spatial solutions	Lack of typological consistency prevents stabilization of layout strategies	Adaptive spatial negotiation through context-specific adjustment of existing structures
Industrial Implementation	Prefabrication and industrialized production ensure efficiency and scalability	Fragmented supply chains and structural constraints limit prefabrication	Hybrid implementation combining prefabricated elements with on-site construction
System Integration	Strong alignment between regulation, design, and production enables coordinated retrofitting	Weak coupling across regulatory, market, and construction systems	Partial and context-dependent integration with loosely coordinated mechanisms

7.3 Reconfiguring Cognitive Construction Mechanisms

Cognitive construction in the Chinese context does not emerge as a coordinated system, but operates as a fragmented and context-dependent process shaped by socio-cultural conditions.

Rather than being stabilized through institutional framing, shared narratives, and repeated experience, cognitive mechanisms remain unevenly formed and weakly connected.

• Methodological Framing

In Sweden, institutions such as HFI established a strong methodological basis for interpreting domestic space, enabling spatial design to be understood through structured and widely recognized evaluative frameworks.

In the Chinese context, such frameworks are weakly developed or absent. Spatial interventions are often perceived as ad hoc modifications rather than outcomes of systematic analysis.

This reflects a weak formation of methodological framing, where spatial configurations are not consistently interpreted through a shared analytical lens.

• Narrative Alignment

The narrative alignment mechanism observed in Sweden links accessibility and functionality to broader narratives of modernity and quality of life.

In contrast, age-friendly adaptations in China are frequently associated with decline, dependency, or loss of independence. This divergence prevents the formation of a coherent value framework that supports acceptance.

Spatial interventions therefore remain misaligned with prevailing social meanings, limiting their perceived desirability.

• Experiential Normalization

In Sweden, repeated exposure to standardized environments enables users to internalize and normalize adapted spatial configurations.

In China, limited and uneven adoption restricts opportunities for sustained interaction. As a result, unfamiliar configurations remain difficult to accept or integrate into

everyday practice.

This condition prevents the formation of stable feedback loops between spatial design and user experience.

These conditions suggest that cognitive construction in China does not operate as a stabilized system, but as a fragmented and context-dependent process.

The effectiveness of cognitive construction depends on the reconfiguration of mechanisms through which spatial environments are interpreted, valued, and experienced.

Table 2: Reconfiguration of Cognitive Construction Mechanisms in the Chinese Context

Mechanism Dimension	**Swedish System (Stabilized Cognitive Logic)**	**Contextual Constraints**	**Mechanism Reconfiguration**
Methodological Framing	Institutionalized frameworks (e.g., HFI) translate spatial design into objective and shared evaluative criteria	Weak formation of shared evaluative mechanisms linking spatial design to user understanding	Context-dependent and partial framing based on localized interpretation
Narrative Alignment	Accessibility embedded in narratives of modernity and quality of life	Misalignment between spatial interventions and prevailing cultural meanings	Incremental and context-sensitive reinterpretation of value frameworks
Experiential Normalization	Repeated exposure stabilizes user perception and acceptance	Limited and uneven adoption restricts experiential familiarity	Fragmented and discontinuous normalization processes
Cognitive Integration	Alignment between institutional framing, social narratives, and user experience stabilizes perception	Weak coupling between perception, meaning, and spatial practice	Partial and unstable integration with dynamic and contingent acceptance

7.4 Toward a Contextualized Dual-Logics Framework

The analysis developed in this chapter suggests that age-friendly retrofitting cannot be understood as a purely technical or design-oriented problem. Instead, it operates as a system of interdependent mechanisms that link material interventions with processes of perception, interpretation, and acceptance.

Building on the analytical framework established in this study, retrofitting is conceptualized through a dual-logics structure, consisting of physical integration and cognitive construction. Physical integration organizes environmental conditions through the coordination of regulatory standards, spatial design, and construction systems, while cognitive construction shapes how these conditions are perceived, valued, and embedded in everyday practice.

The Swedish case demonstrates that the effectiveness of retrofitting depends on the alignment of these two logics within a coordinated system. Standardization, typological consistency, and industrial production enable the formation of physical integration mechanisms, while institutional framing, narrative alignment, and experiential normalization support the stabilization of cognitive construction.

In contrast, the Chinese context illustrates that these mechanisms are not absent, but remain fragmented and unevenly connected. Spatial heterogeneity, institutional fragmentation, and divergent cultural perceptions disrupt the formation and coordination of both physical integration and cognitive construction, limiting the scalability and consistency of retrofitting practices.

From this perspective, the adaptation of age-friendly retrofitting cannot be understood as the transfer of a complete system, but as a process of mechanism reinterpretation. The effectiveness of retrofitting depends on how underlying mechanisms are reconfigured in response to contextual conditions, rather than on the direct application of standardized solutions.

This study contributes to existing research by shifting the analytical focus from design solutions to mechanism formation. It demonstrates that the scalability of retrofitting is not achieved through isolated technical interventions, but through the alignment of regulatory, spatial, industrial, and cognitive processes.

Ultimately, age-friendly retrofitting should be understood as a context-dependent

system in which physical and cognitive dimensions must be jointly configured. This perspective provides a foundation for future research and practice aimed at developing adaptive and scalable retrofitting strategies across diverse socio-spatial contexts.

8 Conclusion

8.1 Summary of Findings

This study set out to investigate the underlying mechanisms that structure age-friendly retrofitting, using Sweden as an analytical case and China as a context for reinterpretation.

In response to RQ1, the analysis demonstrates that Sweden's physical integration logic is structured through the alignment of regulatory codification, standardized spatial design, and industrialized implementation. These mechanisms enable the systematic translation of anthropometric knowledge into scalable spatial solutions, thereby reducing environmental press in a consistent and reproducible manner.

In response to RQ2, the study shows that the effectiveness of these spatial systems is supported by a complementary process of cognitive construction. Through methodological legitimization, narrative alignment, and experiential normalization, accessibility-oriented design is reframed as desirable and embedded within everyday housing practices.

In response to RQ3, the analysis of the Chinese context reveals that these mechanisms cannot be directly transferred due to fragmentation across spatial, institutional, and cognitive dimensions. Instead, adaptation requires a process of reinterpretation, in which physical and cognitive logics are reconfigured to operate under conditions of heterogeneity, partial implementation, and evolving social acceptance.

Taken together, these findings indicate that the effectiveness of age-friendly retrofitting depends not on isolated technical solutions, but on the coordination of underlying mechanisms across multiple dimensions.

8.2 Theoretical Contributions

This study contributes to existing research by developing a mechanism-oriented understanding of age-friendly retrofitting.

First, it conceptualizes retrofitting as a dual-logics system consisting of physical integration and cognitive construction, thereby integrating spatial, technical, and socio-

cultural dimensions within a unified analytical framework.

Second, it shifts the analytical focus from design solutions to mechanism formation. Rather than evaluating individual interventions, the study demonstrates that scalability and effectiveness depend on how regulatory, spatial, industrial, and cognitive processes are aligned and stabilized.

Third, the study advances a context-sensitive perspective on cross-national learning. By interpreting Sweden not as a model to be replicated but as a system to be deconstructed, it shows that adaptation must occur through the reinterpretation and reconfiguration of underlying mechanisms rather than direct transfer.

In this sense, the research provides a structured theoretical foundation for understanding age-friendly retrofitting as a dynamic and context-dependent system.

8.3 Practical Implications

The findings of this study have important implications for architectural and policy practice in the field of age-friendly retrofitting.

From a spatial perspective, the research highlights the importance of moving beyond isolated, add-on interventions toward more integrated approaches that consider the overall configuration of domestic space. Even under conditions of heterogeneity, efforts to improve circulation, functional relationships, and spatial continuity remain critical for enhancing person–environment fit.

From a system perspective, the study suggests that the development of retrofitting practices should focus on building connections between regulatory frameworks, design strategies, and implementation processes. Rather than relying solely on technical standards or policy incentives, effective retrofitting requires coordinated systems that support both physical transformation and user acceptance.

Importantly, the findings also emphasize the role of perception in shaping the success of spatial interventions. Architectural solutions must not only be functionally effective, but also culturally interpretable and socially acceptable.

8.4 Limitations and Future Research

This study has several limitations that should be acknowledged.

First, the research is based on a qualitative comparative approach, relying primarily on document analysis and case-based interpretation. While this enables in-depth exploration of underlying mechanisms, it limits the ability to empirically validate findings across a broader range of cases.

Second, the analysis focuses on Sweden as a single reference case and China as a contextual field, which may not fully capture the diversity of retrofitting practices in other countries or regions.

Third, the study emphasizes mechanism-level analysis rather than detailed design evaluation, which means that specific architectural solutions are not examined in depth.

Future research could extend this work by incorporating empirical case studies of retrofitting projects in China, allowing for a more detailed examination of how proposed mechanism reconfigurations operate in practice. Quantitative approaches could also be introduced to evaluate the impact of different interventions on user outcomes, such as mobility, safety, and quality of life.

In addition, further research could explore how digital tools, smart technologies, and emerging construction systems may influence both physical integration and cognitive construction in age-friendly retrofitting.

References

An, D., Zhao, T., & Zhao, L. (2025). Residential interior age-friendly retrofit design based on the home-based care model. *Construction & Design for Engineering*.

Beijing Municipal People's Government. (2023). *Implementation opinions on improving Beijing's elderly service system*. https://www.beijing.gov.cn/zhengce/gfxwj/202311/t20231101_3293045.html

Bergman, B., Argento, C., & Kågström, J. (2025). Housing adaptation policy for older adults: The role of real estate and municipal actors in Sweden. *Journal of Housing and the Built Environment*. Advance online publication. https://doi.org/10.1007/s10901-025-10255-5

Boverket. (2024). *Om Boverkets nya byggregler (BFS 2024 series)*. https://www.boverket.se/sv/byggande/regler-for-byggande/om-boverkets-nya-byggregler/

Boverket. (2024). *Boverkets föreskrifter om byggnaders tillgänglighet och användbarhet för personer med nedsatt rörelse- eller orienteringsförmåga (BFS 2024:12)*. https://rinfo.boverket.se/BFS2024-12/pdf/BFS2024-12.pdf

Feng, Z., Liu, C., & Li, L. (2025). Towards ageing-in-place in China: The importance of neighbourhood outdoor spaces. *Humanities and Social Sciences Communications, 12*(1). https://doi.org/10.1057/s41599-025-05882-w

Hemmens Forskningsinstitut (HFI). (1950–1970). *Archive materials related to the "God Bostad" exhibitions and kitchen studies*. Stockholm City Museum Archives.

Jonsson, O., Frögren, J., Haak, M., Slaug, B., & Iwarsson, S. (2021). Understanding the wicked problem of providing accessible housing for the ageing population in Sweden. *International Journal of Environmental Research and Public Health, 18*(3), 1169. https://doi.org/10.3390/ijerph18031169

Lawton, M. P., & Nahemow, L. (1973). Ecology and the aging process. In C. Eisdorfer & M. P. Lawton (Eds.), *The psychology of adult development and aging* (pp. 619–674). American Psychological Association. https://doi.org/10.1037/10044-020

Ministry of Civil Affairs of the People's Republic of China. (2020). *Guiding opinions on accelerating the implementation of the home modification project for the elderly*. http://xxgk.mca.gov.cn:8011/gdnps/pc/content.jsp?mtype=1&id=14078

Ministry of Civil Affairs and China National Committee on Ageing. (2024). *Report on the development of the country's work on aging in 2023*. https://english.www.gov.cn/archive/statistics/202408/01/content_WS66a9eacac6d0868f4e8df365.html

National Bureau of Statistics of China. (2023). *Statistical communiqué of the People's Republic of China on the 2023 national economic and social development*. http://www.stats.gov.cn/sj/zxfb/202402/t20240228_1947915.html

Qin, L. (2021). *Research on the practice framework and method of home modification for the elderly* (Doctoral dissertation, Tsinghua University).

Slaug, B., Granbom, M., & Iwarsson, S. (2020). An aging population and an aging housing stock: Housing accessibility problems in typical Swedish dwellings. *Journal of Aging and Environment, 34*(2), 156–174. https://doi.org/10.1080/26892618.2020.1743515

Statens Offentliga Utredningar (SOU). (2015). *Bostäder att bo kvar i – Bygg för gemenskap i tillgänglighetssmarta boendemiljöer (SOU 2015:85)*. https://www.regeringen.se/rattsliga-dokument/statens-offentliga-utredningar/2015/10/sou-201585/

Swedish Standards Institute. (2006). *SS 914 21:2006 – Building design: Housing – Functional design*. SIS.

Wang, Z., Wang, H., & Liu, Z. (2023). Creating a supportive environment for older adults in China. *Journal of Clinical Medicine, 12*(21), 6926. https://doi.org/10.3390/jcm12216926

World Health Organization. (2007). *Global age-friendly cities: A guide*. https://www.who.int/publications/i/item/9789241547307

Wu, S., Fu, Y., & Yang, Z. (2022). Housing condition, health status, and age-friendly housing modification in Europe: The last resort? *Building and Environment, 215*, 108956. https://doi.org/10.1016/j.buildenv.2022.108956

Yang, Y., & Ge, X. (2024). Home modification strategies under the home-based care model. *Art & Design*.

Zheng, Y., Lu, J., & Ding, J. (2024). Responding to life needs: A study on home-based age-friendly retrofit design based on a residential retrofit practice in Beijing. *Time + Architecture*.

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